

# SAFER FIELD RESEARCH IN THE SOCIAL SCIENCES

## A Guide to Human and Digital Security in Hostile Environments

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# 5

## Thinking Systematically about Digital Security

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## Introduction

The digitalization and datafication of everyday life have deeply affected academic research. Social scientists have witnessed the emergence and evolution of new technologically-assisted methodologies and the systematic integration of information and communication technologies (ICT) into their daily practices. There are new ways to generate data, new places to store and catalog this information, new digital publishing formats, and, in consequence, new questions about to whom it will be accessible:

As researchers began carrying more technology with them, they have necessarily learned to take extra care with data before crossing borders and checkpoints. And since maintaining an online presence has become nearly compulsory for academics, regimes and non-state actors have responded by following and, sometimes, intimidating those who voice critiques or challenges to official narratives, raising the concern of censorship. (Parkinson 2017, p. 14)

In recent years, new modes of observation and coercion are emerging that affect researchers, their research participants, and related third parties in the field (Gentile 2013; Glasius et al. 2018; Morgenbesser and Weiss 2018). They are commonly referred to as digital or data-driven surveillance. If surveillance denotes a systematic and focused attention on personal details and behavioral patterns for the sake of control, influence, or entitlement (Lyon 2007, 2014, p. 2), data-driven surveillance implies an additional technical element. According to Marx (2007), it is ‘scrutiny through the use of technical means to extract or create personal or group data, whether from individuals or contexts,’ and includes aerial drones, DNA analysis, data profiling, matching and mining, automatic number plate recognition cameras, CCTV with enhanced definition and predictive functionality, and also imaging and scanning capabilities (Marx 2007, p. 12). As a consequence, sensitive information might fall into the hands of state agents and commercial internet or service providers who seek to intercept research data either independently or in coordination.<sup>1</sup> In hostile research environments, state agencies tap internet infrastructure themselves or compel providers to disclose or hand over desired data about targets, or they obtain this data on a voluntary basis.<sup>2</sup> They also outsource surveillance tasks to contractors and purchase information from data brokers who operate in a complex marketplace for user data (Zuboff 2015).

The proliferation of bilateral and multilateral intelligence- and information-sharing agreements between governments in the frame of the global war on terror, such as the Five Eyes (FVEY) agreement, has facilitated the diffusion of such data across borders (Lyon 2015;

<sup>1</sup>From the perspective of potential data access points, direct surveillance either accesses data from their final destination on the servers of commercial providers or harvests it from the sender device (e.g., through the wiretapping of telephones or the monitoring computers through a local network). By contrast, indirect surveillance intercepts data traffic, i.e., data packets in transit between two ends.

<sup>2</sup>One example for this process is the *PRISM* program, which was revealed through the files leaked by Edward Snowden. In the framework of *PRISM*, several service providers, including *Microsoft*, *Yahoo!*, *Google*, *Facebook*, *Apple*, and *AOL*, allowed the United States government to access their servers.

Reveron 2006; Rosenau 2013).<sup>3</sup> These multilateral agreements often undermine national restrictions on domestic surveillance by enabling governments to solicit information on their country's citizens from the security services of their international partners. The upside to this story is that simultaneously to the proliferation of surveillance practices, a growing community of experts on digital and data security has emerged across the globe. This community has dedicated time and efforts to counteracting the risks of digital surveillance: by creating transparency on the existing threats, by developing and contextualizing technological-security tools, and by providing guidance on their proper use in combination with broader protection practices.

This chapter builds on the work of this community by providing a systematic introduction to the field of digital security for social scientists. It proposes, above all, a context-driven approach to data and communication security and encourages a holistic perspective on research security – online and offline. Rather than outlining a rigid framework for digital security, the chapter focuses on transferable strategies for handling and mitigating data-related risks which transcend one specific medium or research design. These basic digital-security strategies build on pre-existing and intuitive non-digital protection strategies and range from practices to increase the costs of surveillance to ways of minimizing the negative impact of successful digital intrusion attempts.

## Digital threat modeling

As in the process outlined in Chapter 2, the very first step in integrating digital security planning with research designs consists in laying out a threat model. However, notions of 'first' and 'step' can be deceitful: research environments are dynamic, and so is the threat model. During the research process, it is key to revisit and reassess threat models periodically and to scrutinize and adapt the corresponding mitigation tactics and security practices. The choice of the latter should depend heavily on what the threat model looks like. Accordingly, what is needed is a context-driven and procedural approach to data and communication security that can be adapted to the specific needs of researchers and their projects.<sup>4</sup> The following list of questions can serve as a useful heuristic for your threat modeling.

As a first step, people who are not familiar with digital risk assessment procedures can broadly reflect and take notes on *what types of data they actually produce or make use of during their research*. This first step of the threat modeling process is commonly referred to as 'data mapping.' It includes, of course, the information collected and stored as actual research data. But it is more than that: information about the people researchers contact during their field trips, the site and frequency of their interactions, the roads they travel, the volume of transferred data, etc., are just some examples of the data that are generated throughout the research process and that can reveal a lot about a researcher

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<sup>3</sup>For example, data gathered by surveillance programs of the National Security Agency are shared with other governments on the basis of bilateral and multilateral agreements and via joint programs, such as RAMPART-A.

<sup>4</sup>See: <https://securityplanner.org/#/>.

and her/his interlocutors.<sup>5</sup> This data is often referred to as ‘metadata’ (see Box 5.1). When thinking about types of data that should be protected from surveillance, compromise, or confiscation, you should think about these contextual data too. The Citizen Lab provides a ‘Security Planner’ for beginners that can help you to identify areas where you might be vulnerable to digital attacks. This interactive guide is a good way to start the threat modelling, as it walks its users through questions about how they handle private information, specific security concerns, and information about their unique circumstances, and provides respective security recommendations.<sup>6</sup>

### Box 5.1

#### Metadata

From the perspective of assessing risk, the traditional distinction between content and metadata is no longer sound (Goldberg 2017). Metadata is ‘everything about a piece of information, apart from the information itself’ (Electronic Frontier Foundation 2018b). In fieldwork contexts, such ‘metadata’ can become a map of researchers’ aims, their movements and meeting points, and their residence and office (usually those places where their phones remain for longer periods without moving), as well as of their private and professional contacts, their respective social background, and their importance for the research (how frequent and how long are meetings?) (De Montjoye et al. 2013; Mayer and Mutchler 2014). Enabled by the growing processing power of computers, powerful algorithms can be used to process enormous volumes of such metadata to create an accurate profile of individuals.

*Who might be interested in this data, the sources of this data, and its further use? And how and by whom could such data be misused?* Answering these questions is referred to as ‘actor mapping.’ In addition to actors who are supportive of, or opposed to researchers’ aims of protecting their research data and interlocutors (see also the section on ‘Conducting a Context Analysis’ in Chapter 2), it might be useful to list actors separately who are either supportive or indifferent to the researchers’ project, as well as those actors whose stance is yet undetermined and who might become wild cards in the threat model.

*What could these actors possibly do to interfere with field research?* You may think here of threats related to digital technologies as well as their interrelation with the potential physical and socio-psychological threats that you have modeled based on the advice in Chapter 2. For instance, how likely is a sophisticated cyber-attack compared to someone breaking into your field office and stealing your laptop? Depending on the perceived ranking of these different risks, you can decide which protective measures to prioritize; that is,

<sup>5</sup>The ‘Felton Testimony’ provides a good vantage point for interested readers to inform themselves about the possibilities of ICTs to collect metadata and the insights such analysis can generate. See: [www.cs.princeton.edu/~felten/testimony-2013-10-02.pdf](http://www.cs.princeton.edu/~felten/testimony-2013-10-02.pdf).

<sup>6</sup>See: <https://securityplanner.org/>.

whether to focus on digital protection and online security or buy a better office lock or a vault for your files first.

To make an informed choice here, you should ask yourself (and mark down, for instance, in a *table*) *what capabilities and capacities different actors possess and how likely it is that they will use them? Which ones are they more likely to use than others? Local NGOs and international human-rights organizations*, such as Amnesty International, Privacy International, or Reporters Without Borders, are usually good sources for information about the digital capabilities of the state-level and private actors at your fieldwork site. In addition, specialized news outlets, such as the Intercept or Motherboard, often provide valuable insights about digital surveillance technologies used in different parts of the world. Assessing the *likelihood* that such technologies will actually be used against you can be more challenging. Especially in authoritarian states, decision-making processes are often a black box (Frantz 2016; Magee and Doces 2015). Moreover, most surveillance attacks against digital infrastructure go unnoticed by their targets. Hence, researchers can never be certain that reports about cyber-attacks paint a reliable picture of the level of digital surveillance. One way to deal with this uncertainty is to act on the assumption that ‘any online activity, or documents on online servers or in virtual clouds, may be subject to scrutiny’ (Glasius et al. 2018, p. 32). This effectively requires a break with conventional research habits: rather than collecting and processing as much data as possible, researchers who acknowledge the potentially complete transparency of their research projects will need to restrict the data they generate, process, and share to a minimum to reduce exposure. What is more, if a researcher accepts ‘do no harm’ as the ethical imperative of all fieldwork, then the protection of research participants is paramount.

*What could be the consequences if research data were compromised? Who could be affected, and who is most vulnerable?* Lee Ann Fujii, for instance, addressed these questions when faced with the task of anonymizing her research sites and interviewees in Rwanda for her first book (Fujii 2012, p. 721). She imagined it was most likely that only a low-level bureaucrat could be tasked with identifying her respondents, rather than a digital surveillance agent. But as she could not be sure of this and knew about the potential consequences of deanonymization in Rwanda at the time – arbitrary arrest and indefinite detention – she chose to stay on the safe side when trying to protect identities and refrained from recording details about her interviewees in any digital form.<sup>7</sup> Our advice goes in a similar direction: while we do not recommend working from a level of paranoid caution, we believe that putting the safety of research participants at the center of considerations on digital security has consequences for common data collection practices. It ultimately brings up questions about the feasibility of anonymization and informed-consent procedures at academic institutions. If we are uncertain whether we are being surveilled or not, how can we adequately inform our interlocutors about the risks they are consenting to when taking part in our research? Researchers’ transparency about the limits of their assessments is thus key to bringing ethical considerations in line with the risks that arise from uncertainty. This includes a stronger commitment to openness and transparency *vis-à-vis* your prospective research participants. You should reflect honestly upon the restrictions and uncertainties of your research plan,

<sup>7</sup>See her post in section I.2. Research Ethics: Human Subjects and Research Openness: Quality and Transparency Deliberations Forum, October 21, 2016, 10:23 am, [www.qualtd.net/viewtopic.php?f=14&t=116](http://www.qualtd.net/viewtopic.php?f=14&t=116).

and communicate transparently to your interlocutors what you and your peers, including your advisors and supervisors, are really *able and willing to do* to mitigate threats in light of your available resources and ethical commitments.

Above all, this means rethinking resource allocation. There is usually a trade-off between convenience and security. Think of the common and justified fear among many researchers of losing their collected digital data due to a server error, hardware breakdown, failed synchronization, or simply a hasty mouse click. The most convenient way to safeguard against such data loss would be to constantly keep several copies of the collected and processed data at hand. By contrast, from a digital security perspective, reducing the amount of stored sensitive data offers the best protection against data breaches. Retaining and storing multiple duplicate copies of records increases the chance that some of these might fall into the wrong hand in the event of a breach. Researchers concerned with digital security will thus have to steer a middle course that will allow them to restore their research data and resume their work even after primary data loss, while at the same time reducing the risk of compromising data – for instance, by keeping only a few updated and encrypted backups (see Chapter 6), while regularly deleting older copies that are no longer needed. Such ongoing file maintenance greatly reduces the risk of both data loss and security breaches. Devising a sound and holistic, but also viable, security strategy usually entails making such decisions that lie somewhere on the continuum between the two extremes.

Setting up a research team's internal and external communication, data collection, storage, and sharing framework in a secure way also requires time and commitment. It demands constant maintenance and responsiveness to the changing digital landscape. But it pays off in a world where more and more sectors of our lives are dependent upon some digital-data infrastructure (Bertolucci 2013). To stay on top of the digital realities, researchers, jointly with their advisors, supervisors, and team members, need to budget respectively when they compile their grant and fieldwork proposals. If we want to have control over our data, we might have to invest in hardware (e.g., servers), software, and digital-security education. This includes designing secure work flows and assigning responsibilities. However, it can also mean rethinking the need of technology more generally. In some instances, free and open-source software (FOSS) technology may be preferable to expensive high-tech solutions; in others, low- or lower-tech options may potentially be the safest option, from the perspective of data security. Digital security is much more about processes and a holistic understanding of data and how different data types and sources are or can be interlinked, than it is about specific tools or software solutions.

## Box 5.2

### Digital threat modeling

**Main aim:** Identifying data-related threats and assessing the likelihood that they will materialize as well as their potential impact on the research project.

Data mapping:

- What types of data do you produce or make use of during your research?
- What information do you collect, store, or communicate?

- What types of contextual/metadata are being created throughout the research process?
- What would be the consequences if either of these data types were compromised?

Actor mapping:

- Who might be interested in your data, the sources of your data, and in its further use?
- Who might obtain access to your communication via physical or digital means?
- How do these actors view your research project? By whom could your data be misused?
- What are their capabilities and how likely it is that they will use them?
- Which ones are they more likely to use than others?
- Who would be affected by data breaches?

Context analysis:

- How accessible and reliable is the communication infrastructure?
- What software and ICT services do you rely on during your research?
- Does the legal context prohibit or prescribe the use of certain ICT services?
- Does your academic institution prohibit or prescribe the use of certain ICT services?
- What level of digital proficiency do you and/or your research team possess?
- Who else might have access to your computer?

**Outcome:** Creating a matrix of threats, assessing their likelihood and potential impact.

**Example:** Your fieldwork for a European research institute will bring you into contact with sympathizers of the 'Shining Path.' The communist revolutionary guerilla group is classified as a terrorist organization by Peru and the European Union. Its members are regularly targeted by Peruvian military operations. As part of your data mapping you first think about sensitive information sources you might consult to learn about your research context, and which might put you in the spotlight of security services (see the case of the 'Nottingham Two' (Curtis and Hodgson 2008)). Then you consider the primary data that you might collect (such as testimonies, interviews, archival material) and the secondary data you produce (memos, field notes, emails, text messages) and how you are planning to communicate and store it. In addition, you list the metadata created during your field trips, including the virtual networks emerging from your meetings with different interlocutors and the geo-spatial data generated by your travels. Finally, you think about your use of social media and how it increases your digital exposure.

During your actor mapping, you realize that both local law enforcement as well as European security services might be interested in your data. International and local human rights organizations and data activist collectives provide you with a tentative picture of their respective capabilities: Peruvian security services would most likely attempt to seize relevant data while you are in the field, through either digital or physical surveillance. However, they might continue to monitor your work even after the conclusion of fieldwork in expectation of future benefits. By contrast, European intelligence agencies are most likely to obtain your data via digital surveillance.

Aside from law enforcement, your project might face opposition from local civil society and peasant populations who have suffered from the violence deployed by the group. For your

*(Continued)*

context analysis, you collect information from local news outlets and specialized news portals, which suggest that Peruvians enjoy unrestricted internet access and can legally use VPNs, thus allowing you to employ specialized software to circumvent censorship and minimize the risk of digital surveillance. As part of your context analysis, you also realize that you have so far relied on insecure software for collecting and processing research data, which might have exposed your research project to state surveillance.

## Thinking of data holistically

You can think of data as all the information that either informs or is created as a product or by-product of your research. This data can be anything, from handwritten notes that will be processed through digital means, to digital records in text, audio, or video format. During fieldwork, however, data is not only the information that researchers proactively collect, e.g., through interviews or participant observation; it also includes the traces left behind when navigating the academic terrain before, during, and after fieldwork. When mapping actors and sketching threat models related to discrete research projects, it is useful to think of degrees of sensitivity of data and to map the data that will be handled and generated through both fieldwork and desk research. During every research process, evidently diverse types of data are collected, communicated, processed, and stored, by the researchers, their affiliated institution, and third parties. To minimize the risks for interlocutors, academic researchers usually seek to keep this data – as well as the details of how it is collected, handled, and stored – secret or, at least, inaccessible to third parties. Many researchers are unaware, however, that their ability to maintain confidentiality is ultimately contingent on the hardware and software they use to handle research data, as well as the email services and communication platforms they resort to on a daily basis. In fact, misunderstandings of the ‘lives’ of digital data are common.

While an ideal research process, which involves different stages of data manipulation, follows a somewhat linear pattern (starting with data collection and ending with a publication), it is important to bear in mind that not all the data-handling processes during a typical research project occur sequentially. When data is digital, they almost never do. Instead, processes of data handling overlap, run in parallel, and plug into each other at different points in time. For instance, an act of digital communication (e.g., a *Skype* call, an email message, or a *Whatsapp* chat) is always an act of data storage too. Researchers usually know who they communicate with – respondents, interlocutors, fellow researchers, or supervisors. However, they rarely know the third-party actors involved in storing and processing the communicated data. These actors include local internet providers, software distributors, or surveillance companies, and can be of questionable trustworthiness. The implication of this triangular character of digital research communication for the security of researchers and their respondents is that there are other research-related data than the consciously and proactively collected field data (field notes, interviews, surveys, and the like) that can be considered sensitive and may require protection: the contextual personal data created by researchers in their everyday interactions through and with information and communications technology (ICT). These data can be roughly divided into three categories: user-generated (volunteered) data, observed data, and inferred data.

## Box 5.3

## Data types

| Data types          | Description                                                                                                                                                                       | Examples                                                                                                                                                              |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User-generated data | Data shared online by individuals on websites where registration or self-identification is required for a service                                                                 | Public and private social-media data, survey data, content registration forms, data recorded by service membership cards and loyalty programs, and registration forms |
| Observed data       | Data created and gathered during transactions between individuals as customers and companies                                                                                      | Credit-card transactions, telephone call records, purchase history at retailers, location data from mobile phones, browser history                                    |
| Inferred data       | The output of data aggregation and processing by technological means (data analysis, especially data mining using machine learning), including the cross-referencing of data sets | Credit scores, predictions of preferences, purchase intents, inferred sexual or political orientation                                                                 |

The first category (user-generated data) refers to the data you share voluntarily, for instance, by entering information on websites upon registration or when you log into your account for certain online services. Volunteered data is often associated with social media, where millions of individuals, including researchers, share both their private interests and professional concerns. However, it is also collected by most mobile phone applications, via surveys, content registration forms, service membership cards and other loyalty programs, and registration forms. Distinct from this user-generated (i.e., individually provided) data is 'observed data.' According to the World Economic Forum (2011, p. 7), these data include credit-card transactions, telephone call records, retail purchase histories, and location data from mobile phones. It is usually created and gathered during transactions between individuals as customers and companies. Finally, inferred data denotes what is commonly referred to as big data – the output of data aggregation and processing by technological means (Lyon 2014, p. 2). The production of inferred data relies on the collection, combination, and processing of data with the help of computer algorithms. These processes are conducted by both private companies and governments and includes, among others, credit scores, predictions of preferences, interests and purchase intents, as well as the extrapolation of individual or group behavior. Of course, these inferred data might be wrong, but the consequences for those identified through it – for instance, as your research participants or informants – are likely to be real. What is more, when new data is created through statistical and computational means, the inferred data is not considered personal data and is thus not protected under data-protection legislation (Miljanovic 2015b).

All three data types can be valuable assets for those with an interest in surveilling and restricting critical researchers: the first two data types can be associated with the risk of interception (e.g., through targeted or mass surveillance) and the third can be related to the capabilities of state-level actors to either produce it, purchase it, or request and obtain

it from private companies. In fact, the frequently changing terms of service (TOS)<sup>8</sup> and data-processing consent forms of many popular ICT providers, such as *Google*, *Facebook*, and *Microsoft*, are often so lengthy and indecipherable that their readers are unaware of what they are consenting to – for example, the resale of profile data to commercial data brokers (Beckett 2012; Gutwirth and Hildebrandt 2010; Hildebrandt and Gutwirth 2008; Hoofnagle 2004; Miljanovic 2015a; Turow 2012). These data brokers are companies that profit from the sale of user-related data that circulate online, are found in public databases, or are bcollected from online and offline platforms. They compete in a growing market where some companies, such as *HackingTeam*, sell data acquired via software exploits or hardware backdoors; others offer analytics software and other tools to collect observed and user-generated data; and yet others trade the data inferred from the users of commercial digital products and services.

Zuboff (2015) has tellingly referred to this monetarization of user data as a new form of ‘surveillance capitalism.’ Unsurprisingly, *Google* is currently the most powerful entity in this economy. It centralizes and shares vast amounts of detailed user data (Goodman and Powles 2016; Solnit 2013).<sup>9</sup> Its commercial practices of collecting, processing and sharing data, including with other companies and government agencies, are most commonly associated with the notion of ‘corporate surveillance.’ However, government bodies or their proxies are involved in this market too, both as collectors and as buyers of user data (Rosen 2012; Schneier 2013) – probably including those states where you plan to conduct your fieldwork. They might either be directly interested in the field habits of researchers and their interaction patterns or draw on this context information to identify their field contacts.

Cross-referencing, integration, and the merging of different data types available for sale make it possible to obtain information and create profiles<sup>10 11</sup> of researchers, their contacts, and their activities indirectly (Lyon 2014). Profiles are composed of various combinations of data points. A small number of individual pieces of data, once combined, may suffice to identify an individual – even before the fieldwork has started. Given these potentials of cross-referencing, you should be aware that seemingly innocuous and isolated search terms

<sup>8</sup>Users can get an intelligible overview of the terms of service of popular service providers at <https://tosdr.org/>.

<sup>9</sup>Google’s (Alphabet’s) extensive and quasi-monopolistic reach is enabled by myriad services and products, such as *AdSense*, *Google Maps*, the *Android* platform, and *Gmail*, all of which bring monetizable user data to *Google*, as well as acquisitions of and partnerships with other companies producing hardware or software through which user data are gathered. See: <https://laloma.info/en/projects/googleempire>.

<sup>10</sup>For example, *Spokeo*, a United States-based data broker, sells profiles containing a person’s full name, physical address, phone number, age range, email address, hobbies, photos, ethnicity, religion, and social network activity. *Nugg.ad*, a German analytics company, builds user profiles that include the user’s web-browsing history.

<sup>11</sup>For example, *Spokeo*, a United States-based data broker, sells profiles containing a person’s full name, physical address, phone number, age range, e-mail address, hobbies, photos, ethnicity, religion, and social network activity. *Nugg.ad*, a German analytics company, builds user profiles that include the user’s web-browsing history.

or pieces of information shared digitally might reveal a lot about your own identity as well as that of your potential interlocutors (Gallagher 2013).

An innocuous credit-card purchase or a 'like' on *Facebook* can become significant, even years later, as one element in a larger profile of you and your research participants.<sup>12</sup> Even people whom you have not been in direct contact with could potentially be profiled as a likely future interview partner based on features they share with your other interlocutors.

## No silver bullet

Now, how can you protect yourself and your data against these practices and its consequences? The most prevalent direct demand encountered by digital security trainers is the one for a 'silver bullet' to obliterate digital-security concerns. That is, a list of technical tools and apps that, combined, somehow mitigate all digital risks once installed. This is also how we started to conceive this handbook in the first place. However, in empirical reality, the list of secure ICT is in constant flux. New risk vectors emerge daily, new applications enter the market, and surveillance is tailored to the routines of their targets. Accordingly, digital-security trainers advise against trusting any software that advertises itself as 'spy-safe.' A tool whose developers emphasize its limitations is more trustworthy than one that does not acknowledge it is not perfect. In the end, no tool is ultimately attack-proof. This is because the security provided by some technical tools at your disposal can be offset by human error. There are good practices for using some technological tools for research, and there are bad ones. And even if you are technically versed and well informed of the risks of certain ICT, your research participants might not be and thus offset your precautions and security measures (vice versa, research participants can also be compromised by careless researchers). Furthermore, sophisticated combinations of traditional offline surveillance methods and refined data forensics may limit or even reverse the effect of a specific communication tool on its own.

Most importantly, adequate data protection strategies are highly context-dependent. For a researcher of illicit smuggling routes from Turkey to Egypt, who took part in one of our workshops, the idea of a sophisticated digital surveillance effort seemed highly unlikely. ICT-affiliated risks related much more to the routines of local police officials who, at most, would randomly *Google* individuals held up at checkpoints and, at most, screen their call history. In turn, for another researcher who had conducted interviews with former ISIS supporters in Iraq, the possibility that research data could be digitally accessed by an interested high-capacity intelligence agency was a plausible scenario, entailing the need for an entirely different approach to data security.

The safety provided by software solutions is equally dependent on the site of your field research: secure browsing technologies that might mitigate much of the risk associated with online information queries in one country may raise red flags in others. For instance,

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<sup>12</sup>For a recent example of the use of profiles based on social-media data for the purpose of voter influencing, see *The Guardian's* dossier on the *Cambridge Analytica* files at [www.theguardian.com/news/series/cambridge-analytica-files/](http://www.theguardian.com/news/series/cambridge-analytica-files/).

routing internet traffic through the *Tor* network and using the *Firefox-based Tor Browser* (see Chapter 6) has proven a relatively secure way of researching information and circumventing censorship online.<sup>13</sup> However, it is illegal in several authoritarian contexts, including China, Turkey, Iraq, the United Arab Emirates, Belarus, Oman, and Russia. Recently, several countries have also cracked down on mobile messaging services. In Turkey, the messaging app *ByLock* has been associated with the Gülen movement, which authorities hold responsible for the 2016 coup attempt, thus placing its users in danger. In Iran, authorities announced in March 2018 the intention to prohibit the *Telegram* app, which was previously used to organize oppositional protests. Having suspicious software installed on a laptop or a questionable app on your mobile phone might raise the suspicion of security agents and thus provide a welcome pretext for further examination.

In a similar vein, relatively safe direct-messaging apps with end-to-end encryption may raise the profile of researchers in countries where only a few people (usually human-rights defenders, human-rights lawyers, or journalists) use these applications and thus attract more attention than needed. This is particularly true for messaging apps such as *Wire* or *Signal*,<sup>14</sup> which have gained popularity among journalists and activists in the aftermath of the Edward Snowden publications and have been recommended by prominent digital-privacy advocates (Kazansk et al. 2016) due to their advantages over conventional email, phone, and Skype conversations. In sum, when designing a digital strategy for your fieldwork, it is wise to consider the political context of every digital technology that you intend to use.

The same holds true for the cultural context of field research: in many situations of personal interaction between you and your interlocutors, the insistence on unconventional, albeit more secure, technological tools can introduce an element of strain. Researchers with field experience can confirm that the reliance on low-key technological tools often makes a difference in terms of interview dynamics. It makes a difference in terms of openness and trust, whether there is an audio-recording device on the table or merely a pen and a notebook. The most open conversations usually take place in a context that differs little from daily routines. Coincidentally, recording interviews in handwriting or taking paper notes may sometimes not only be more viable and efficient as data-collection practices, but also more secure with a view to digital risks: if handwritten notes and transcripts are digitalized only shortly before ‘offloading’ (see Chapter 7), the window during which sensitive information might be compromised by digital intrusion is minimized. In addition, researchers can devise containment strategies before entering the field. They can also think about how to sensitize their research participants to the vulnerabilities of certain means of communication, such as video calls via *Skype* or casual chats via *Facebook Messenger* – tools that have proven to be highly penetrable by a wide range of actors, from criminals to corporate operatives collaborating with law enforcement agencies, to state security institutions themselves. You can and certainly should alert your research participants to these risks. However, in many cases it is wise to tread lightly to avoid appearing paranoid or alienating your interlocutors by criticizing practices they are used to in their daily lives. In addition,

<sup>13</sup>See: [www.torproject.org/](http://www.torproject.org/).

<sup>14</sup>See: <https://wire.com/> and <https://signal.org/>.

often the method of communication is not up to the researcher. 'In the field you often cant choose the means of communicating with your interlocutors. Usually they do. And usually you are happy that they want to talk to you at all!'<sup>15</sup>

## Strategies for handling digital risk

Many of our peers and colleagues have a high degree of stress about digital-security risks. Feeling unable to come up with the skills required to make your research safer can be paralysing. Many of the authors of this volume have experienced this frustration and mental overload. However, we also share the conviction that complexity need not lead to resignation. When devising digital-protection strategies, you and your supervisors do not need to start from scratch. Invariably, you have already developed *ad-hoc* deterrence and protection strategies to handle risks and threats in the past, and these can be adapted to cover some digital risks too. Often a good start is to revisit prior behavior in light of new potential digital threats in the field, and to exclude or modify those strategies that are particularly vulnerable to digital threats while reinforcing those that still seem effective. In turn, when designing new strategies for dealing with digital threats, these should aim at satisfying the same 'RASER' criteria as their non-digital counterparts: Responsiveness, Adaptability, Sustainability, Effectiveness, and Reversibility.

## Slipping through the net

Most social scientists will not and probably cannot enter the 'narrow realm' of digital-security professionals (Zuboff 2015, p. 85). A maximalist approach to digital security – e.g., avoiding several popular co-working platforms and publishing portals for digital-security concerns or the encryption of all of one's communication, field notes, location data, and analyses – can be counterproductive to academic workflows and cooperative research (while potentially also increasing the attention of authorities; UNESCO and Reporters Without Borders 2016, p. 70). Accordingly, researchers carefully need to strike a balance between their capabilities, time, and resources; the exigencies of their tasks; and the assessed risks of their fieldwork. In many cases, a strategy of slipping through the net has proven useful to mitigate and reduce the risks of exposure. The idea of this approach is to stay under the radar of security agencies or at least not to raise too many alarms with your research. If you are deliberately targeted by a security agency there is little chance of staying hidden. However, sophisticated targeted surveillance is costly and requires a substantial amount of resources. The decision to deploy such surveillance technologies against individuals is usually not made at the lower ranks of security agencies. In many research contexts, government agencies will rely first on cheaper mass surveillance to identify and pre-select targets that seem worthy of such surveillance measures. The goal of slipping through the net is not to disappear completely from the radar (which is hardly feasible), but to influence their cost-benefit calculations by reducing the perceived benefits.

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<sup>15</sup>Conversation with Francesco Strazzari at SAFEResearch Workshop, Athens 2018.

This entails engaging with practices that do not promise complete privacy, but offer selective privacy. Van der Velden has referred to these as ‘practices that constitute “shadows,” moments of “hiding in plain sight”’ (Van der Velden 2015, p. 192). Essentially, such ‘hiding in plain sight’ means concealing only your most sensitive activities while exposing non-critical, or at least tolerable, activities to the authorities. The experience of many seasoned field researchers of conflict environments shows that, if you carefully curate your digital presence, you can slip relatively easily past many of the nets spun by governments and security services to detect subversive activities and entrap individuals deemed a threat. Many of these rely on rather superficial scans of online activities, social-media and web profiles, including publication lists from researchers’ home institutions or in popular online repositories. If you are not willing (or able, for research purposes) to shut down completely or limit your activities on *Twitter*, *Instagram*, *Facebook*, and the like, you may use this, to a certain extent, to your advantage: before departing for fieldwork, ask yourself which critical data you want to protect most and prioritize accordingly. For instance, you might continue your daily web activities on social media to simulate normality, while refraining only from sharing research-related news.<sup>16</sup> Alternatively, you might deliberately post only such information on social media which understates your ambitions or frames your research interest and goals in a less sensitive way. In either case, the careful curation of your digital presence may not only avert authorities’ suspicion, but also contribute significantly to the construction of a credible narrative about your work. This may come in handy should you get in trouble with law enforcement, for example, by enabling plausible deniability.

Other measures that can help you maintain a low profile range from retrospective content curation on social media, including the selective deletion of previous posts, contacts, and messages,<sup>17</sup> to the modification of short bios and current research topics on department websites, to the planting and dissemination of fabricated interview transcripts or research proposals on your hard drive for the unlikely case of confiscation. Again, which of these measures is best suited to help you slip through the net is highly context-specific. However, what most precautionary measures for hiding in plain sight have in common is that they cannot be implemented *ex post*, when already in the field. Repressive regimes usually keep an eye on the established foreign-researcher community, monitoring their activities continuously and often from long before they ever set foot on their soil. Sometimes even foreign embassies function as hubs for this intelligence collection. Since 2014, for instance, activists and scholars of Egypt increasingly reported encounters with embassy staff in European capitals, who confronted them with detailed reports about their

<sup>16</sup>Communicating sensitive data elsewhere while maintaining a social-media presence is a so-called ‘compartmentalization tactic.’ Other compartmentalization tactics are described in more detail in Chapter 6.

<sup>17</sup>While *Facebook* does not actually delete any data, the removal of sensitive posts or images from the timeline can decrease the chance that a superficial scan of social-media activities raises a red flag. *Facebook* now enables keyword searches of past postings by users, making it easier to remove submissions mentioning specific countries, research, or sensitive topics. Furthermore, several browser extensions, so-called ‘timeline cleaners’ that directly access your activity log, allow you to bulk delete posts and likes for specific timeframes.

activities and publications. In some of these cases, the records presented reached years into the past, thus documenting the sustained efforts of Egyptian security services to hold critical academics in check. These security incidents exemplify how curating your public profile is not a short-term proposition. Scholars should keep in mind that there might be an ongoing file on their work that cannot be easily erased. A general word of advice regarding your social-media footprint is thus to assume that everything that you write is public and can potentially be stored, analyzed, and used for profiling (Sheline 2016, p. 6).

## Increasing the costs

It can be quite difficult to ascertain whether you have slipped through or just cannot see the net. Therefore, it is wise to think about deterrence strategies that complement your attempts to fly under the radar during your fieldwork. Deterrence strategies also focus on influencing the cost–benefit calculations of those with an interest in your personal and research data. However, in contrast to the approach presented above, their intent is not to decrease the perceived benefits of putting you under surveillance and breaking into your devices or the databases and commercial platforms where your data are stored. Instead, they tackle the costs of such measures and aim to persuade potential attackers that it is simply not worth spending the required assets to access these data (Kazansky 2015, pp. 13–14). The premise of this approach is, of course, that surveillance is costly – a fact that holds true today but is bound to change as the availability of cheap surveillance technologies increases (Rosenbach, Poitras, and Stark 2013). For now, however, many types of attacks are still expensive, especially when the targeted researcher has implemented some digital-protection measures. For instance, thus far, for a security agency, the most effective way to eavesdrop on your end-to-end encrypted communication is to install malware on your smartphone by exploiting weak spots in its operating software – so-called ‘exploits’ – which have not yet been fixed with updates by the software developers. Information about these vulnerabilities can be immensely expensive, with working exploits for the *iPad* and *iPhone* having recently reached a market value of up to US\$1.5 million (Franceschi-Bicchieri 2017). The training and deployment of desk officers who sift through the data collected via malware adds to these initial acquisition costs.

Once again, however, your ability to deter security services from attempts to gain access to your data heavily depends on your research context. Some regimes certainly have these capabilities and manpower and several Western countries command intelligence apparatuses that even stockpile exploits for future occasions. Others lack the required resources themselves. Outsourcing digital surveillance to private contractors is equally costly (Lyon 2014, pp. 2–3).<sup>18</sup> Many governments therefore rely on substantially cheaper – but also easier to spot – phishing and malware attacks to gain access to researchers’ and activists’ data

<sup>18</sup>As reported by *The Washington Post*, in recent years about one in four US intelligence workers has been a contractor, and roughly 70% of the entire US intelligence budget has gone to private firms, including companies such as Booz Allen Hamilton, the former employer of whistle-blower Edward Snowden. See [www.washingtonpost.com/business/nsa-leaks-put-focus-on-intelligence-apparatus-reliance-on-outside-contractors/2013/06/10/e940c4ba-d20e-11e2-9f1a-1a7cdee20287\\_story.html](http://www.washingtonpost.com/business/nsa-leaks-put-focus-on-intelligence-apparatus-reliance-on-outside-contractors/2013/06/10/e940c4ba-d20e-11e2-9f1a-1a7cdee20287_story.html).

(see Chapter 6). The Syrian Assad regime, for instance, has bought and employed advanced surveillance equipment from US companies to intercept the communications of dissidents and rebel groups. Spyware deployed through phishing attacks, however, has played a crucial role in the regime's effort to identify and expose critical journalists and field researchers on Syrian territory (Xynou and Al-Khatib 2015). Even resource-rich states often rely on phishing rather than sophisticated hacking. For example, in 2017, the Toronto-based research group *Citizen Lab* uncovered the involvement of the Israeli surveillance company NSO Group in a phishing scheme in Mexico: activists and journalists were targeted by false *WhatsApp* messages containing spyware in an attempt to silence government opposition. The same spyware was used also in the United Arab Emirates to target the Emirati award-winning human-rights defender Ahmed Mansoor, and most recently in Saudi Arabia to target the international staff of Amnesty International.<sup>19</sup> Expensive exploits are thus hardly needed when data can be purchased cheaply on dark markets for data or obtained through phishing. The upside of this is that there are many tools and techniques available to researchers for fending off such attacks. A significant share of digital-security threats are not highly sophisticated. Raising the intrusion costs by building up a first line of defence against such attacks is thus a relatively easy strategy.

## Minimizing the benefits

Preventive strategies, such as those presented above, are most efficient in dealing with potential digital harm. However, they do not always suffice. Accordingly, it is equally important to think about contingency plans and prepare for the case that you do not manage to remain beneath the radar and your deterrence strategy fails. Reactive mitigation strategies, therefore, intend to reduce the negative impact of your data or devices falling, for instance, into the hands of hackers after your computer has been infected with malware, or of data forensics experts after it has been seized by police or security services.<sup>20 21</sup> Reactive mitigation strategies are designed for precisely these cases and mostly aim at protecting your most

<sup>19</sup>For more information on the attacks, see: [www.amnesty.org/en/latest/news/2018/08/staff-targeted-with-malicious-spyware/](http://www.amnesty.org/en/latest/news/2018/08/staff-targeted-with-malicious-spyware/).

<sup>20</sup>Several guides can prepare researchers for having their mobile phone or laptop seized by law enforcement authorities. For a beginner's guide, see the basic prevention tips released by the Freedom of the Press Foundation at <https://freedom.press/training/preventative-mobile-security-tips-activists/>. For protest researchers, Martin Shelton has compiled helpful guidelines at <https://medium.com/@spacefactor/@mshelton/secure-journalism-at-protests-a435b2d6398b>. However, researchers should be aware that no guidebook or preparation can protect against the scenario of being coerced by law enforcement or border police into giving away passwords and pseudonyms.

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sensitive data by limiting their usability and accessibility in the case of their breach. Many seasoned researchers intuitively rely on such mitigation strategies already when dealing with offline data, for instance, by keeping only partial notes of interviews in their personal notebooks, by anonymizing the names of interviewees, and by obfuscating the timing and location of certain meetings. Another prominent strategy of data protection in hostile contexts is the hidden or compartmentalized storage of analog data, such as filled-out questionnaires or interview transcripts in separate places. Whether analog or digital, the basic idea behind such reactive protection strategies has always been to reduce the value of individual data fragments in case they fall into the wrong hands. A colleague who has been researching rural insurgent communities in the Philippines recounts, for instance, how she used to bury her recording device and interview tapes in two different hidden locations during her field stays, to decrease the likelihood of compromising sensitive data in case of a raid by counter-insurgency forces. Another colleague described how he always carried a lighter and a can of white gas on trips in the mountainous west of Colombia in order to be able to char his field notes if needed. Many of these practices can be translated to digital devices, too, thereby relieving you of the need to bury your field notes and thus potentially lose them to bad weather or bad memory.<sup>22</sup>

## Conclusion

To sensibly implement these strategies, researchers need to use the tactics that best fit their circumstances. Digital-security trainers and data activists often stress the need to design strategies that are both practical and take into account the protection requirements of each specific research process. Furthermore, an inherent part of the threat-modeling process and the design of protection strategies is about an awareness of the tradeoffs between convenience and security. In the end, digital-security strategies ultimately must remain realistic. It is thus key that you and your supervisors ask yourselves to what lengths you are willing to go when it comes to implementing digital-security measures. Inaction is usually the riskiest path. With the help of this handbook and the descriptions in the next chapter, researchers can increase their sensitivity and build up their IT-proficiency by gradually adding new layers of protection measures to their research framework. Some of these layers can be implemented quickly and can offer huge protection with very little effort. However, especially some of the higher-security practices and more sophisticated tools require time, efforts, and eventually training that must be factored into the planning of your field trip. Integrating digital-security management in our way of working is a process that takes time, participation, and commitment. But this process is also an empowering one. As Zuboff notes, ‘exercising one’s right to privacy produces choice, and one can choose to keep something secret or to share it’ (Zuboff 2015, pp. 82–83).

It is useful to seek assistance beyond disciplinary boundaries throughout this process (Dencik, Hintz, and Cable 2016; Milan and Hintz 2013). Learning happens best in conversation and collaboration with those who have long years of expertise when it comes to digital-security measures. Security trainers, for instance, can help you prepare for more

<sup>22</sup>Some secure methods of disposing of sensitive field notes are discussed in the next chapter.

sophisticated digital attacks. Human-rights defenders, digital-activist groups, and strategic litigation NGOs, including Amnesty International, the Electronic Frontier Foundation, Frontline Defenders, Privacy International, Security Without Borders, and Tactical Tech, are out there for support and advice. This requires researchers, and data activists, to rethink the boundaries between academics and non-academics and to embrace a cultural shift toward stronger collaboration across disciplines and sectors of knowledge production.

## Guiding questions

### Questions for researchers

- Do you have a good overview about the field site or socio-political context of your research?
- What needs arise from this context regarding digital protection measures?
- Did your risk assessment include a digital threat modeling (see also Chapter 2)?
- Does your threat model cover the different types of data (volunteered, observed, inferred) that may require protection?
- Have you reflected on the feasibility of your envisioned anonymization and informed-consent procedures in light of their interrelation?
- Have you thought about mapping systematically which data you will collect during your research? Which of these data need to be protected?
- Do you have a good overview about both friendly and possibly hostile actors in the field?
- Which of these actors might be interested in your research data or your sources (before, during, and after fieldwork)?
- What are their capacities (assume high capacities as a starting point)? How likely is it that they will use these capacities?
- Have you consulted with colleagues and supervisors about your assessments? Do you have any information about similar cases or prior security incidents? What can your peers and colleagues tell you about this?
- Have you cross-checked your assessment of potential threats with digital security experts (or websites)?
- Have you thought about the potential consequences of data falling into the wrong hands?
- What threats could arise from this – to you and to your research participants?
- What are the consequences of these threats, if carried out?

### Questions for advisors/supervisors

- Have you talked to your advisee about their digital exposure and about data-related security risks?
- Do you possess sufficient information to correctly advise researchers of the data-related risks of their fieldwork?
- Have you cross-checked your assessment of potential threats with other colleagues and with digital security experts (or websites)?
- Have you ensured that the researchers under your supervision possess sufficient IT proficiency to not endanger themselves and their research participants?
- Have you thought about specific support structures you could provide to enhance the digital literacy of researchers under your supervision?

# 6

## Tools and Tactics for Data Protection before, during, and after Field Work

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## Introduction

The question of how the proliferation of digital research practices affects our physical security and socio-psychological well-being has attracted increasing attention throughout the last years. Above all, authors from the field of critical data and community studies (Andrejevic 2014; boyd and Crawford 2012; Dalton, Taylor, and Thatcher 2016; Dencik, Hintz, and Cable 2016; Tufekci 2014; Van Dijck 2014; Wissenbach 2018) have engaged with the question of how datafication has enabled the emergence of new technologies of domination. Recently, scholars and academic institutions have also noted the need for more practical guidance regarding digital security and data protection (Glasius et al. 2018; Grimm and Saliba 2017; Milan and Trere 2017; Parkinson 2017). This chapter heeds these calls. It provides a comprehensive guide that brings together the advice of data activists and digital-security experts, on the one hand, and the experiences of seasoned field researchers on the other. Unlike other available digital-security guides, it is specifically tailored to the needs of academic researchers and covers the precautions that you can take to protect your communication and data throughout different stages of the research process. The technical advice and examples for good practices intend to provide an entry point and encourage further engagement with the field of digital security among social scientists.

As conceived in this chapter, digital risks arise from researchers' everyday practices with regards to three ideal-typical dimensions: communication, data generation/processing, and data storing. These dimensions intersect. Academic research is a cyclical and repetitive process, rather than a linear one – consider aspects such as pilot phases, the adaptation of research designs for further work or satellite projects, or the reuse of interview data for different publications. Each dimension is prone to specific kinds of digital threats. They relate to the complex layers of communication that happen throughout the research process, between researchers and their supervisors, between members of the same research project, and between field researchers and their interlocutors. And they depend on the methods by which research data are recorded, processed, and stored.

## Communication process

One risk dimension involves the interception and logging of communication between researchers and their respondents, as well as between researchers and/or interlocutors and third parties. This can be by the wider research network, media, state bodies, commercial service providers, or other actors interested in the data. In order to minimize the risk, it is essential to thoroughly discuss and identify the communication processes a research project will use in preparation for the research activity. This includes internal communication flows, such as between researchers and their team members or supervisors before, during, and after fieldwork, or between researchers and their administrative bodies. It is also crucial to keep revisiting implemented practices and procedures throughout the project and to update the communication strategy wherever new potential threats arise. Generally, any time the exchange of information takes place, there are direct or indirect risk factors to mitigate. The subsequent sections of this chapter should help you, prior to the implementation of your research projects, to reflect on whether your communication routines are potentially vulnerable to attacks and to form an informed opinion on the following questions:

- When you share your data, how do you do it and with whom?
- Have you evaluated different communication channels for collecting data and mapped the different tools that you typically use your daily research routines?
- What are their weaknesses and are there alternatives?
- Have you listed all the actors who might obtain access to this communication via physical or digital means (interception, tapping, buying, obtaining on the basis of a prior sharing agreement, requesting authoritatively)?
- Are you using an encrypted connection and secure messenger applications? If not, should you consider using encryption for your email communication or rely on better communication tools?

## Data collection and processing

Risks also arise from the ways data are collected and processed, in the shape of data analysis, and the sharing and reporting of findings, both in the field and remotely. Data are collected not only by you as a researcher, but also by other actors, with or without your awareness. Increasingly, information is also retrieved through mailing list and online forum analyses, or the generation of large-N social-media data sets (Twitter feeds, etc.). As those instances are increasingly also subject to our research, this bears further risks that can affect us, our interlocutors, and third parties. One layer of risk is also our physical navigation in the field and in how far we can be traced through the technology we carry and use, our personal interactions, or the methodologies by which we collect and offload our research data. Further risks arise from the way we are handling our data. Whereas we might have thought about anonymizing our interviews, through triangulating data sets related to your interlocutors, to you, and your movements and practices, critical information about your interlocutors can be assembled and put people at risk. Note: effective anonymization requires more than changing the names of your interlocutors or removing personal identifiers from your data sets and reports. In order to take precautions, ask yourself:

- What software and ICT services do you rely on when processing your data? Are there more secure alternatives?
- Are you processing your research data with the help of software or services that encrypt the data input right away?
- How do you anonymize your data?

## Data storage

Data storage is almost always a key point for potential surveillance that can affect all stages of research: pre-field, the field stay itself, and the period after your fieldwork. In our daily routines, we usually do not ask ourselves explicitly where all the documents we create are stored. Nor do we usually question the easier ways of sharing documents with our peers (e.g., through cloud software). When it comes to sensitive field research, however, it is key to consider who might have access to these data. A thorough threat modeling should take place to determine the extent to which the data stored on any device on- or offline could pose a threat to any participant, institution, collaborator, researcher,

or other entities who may or may not be directly or indirectly involved with the research. This includes interlocutors, fixers, facilitators, and private contacts. Data storage does not refer only to your interview recordings, transcripts, or field notes. It also refers to things such as planning documents, planning conversations via email, or messenger applications. Researchers keep these files stored in many different ways. Furthermore, it is important to consider that data can be compromised in the process of it being stored, as well as while it is stored. Moreover, even if a single piece of intercepted information might not reveal concrete details about you or your interlocutors, triangulation and the merging of data sources can reveal information that can put you or others at risk. Therefore, how data is stored matters as much as where data is stored. This exemplifies o In order to devise security practices against unauthorized access, breaches of confidentiality, or data theft, you may think about a safe data-storage location, such as known and encrypted servers, and a safe mode of storage.

- Is your stored data, including all backups and copies of collected data, anonymized and encrypted?
- Do you know the location and the owner of the server on which you are storing your data?
- Who has access to that server? Do you trust these parties?
- Are you storing any data sets which could reveal identities or patterns when triangulated in separate and secure spaces?
- Do you have any sensitive research data stored on your computer (even temporarily)?
- Does anyone else ever have access to your computer?

## Mitigation tactics

The following sections on ways of communicating safely and on secure practices for processing and storing research data introduce some of the tactics that have been employed by, and proven useful to, journalists and activists. The objective of these tactics is to reduce researchers' exposure, to increase their capacities for resilience, and to minimize the potential points for attacks on their work. These technological tools are not default protectors. They are a means to an end, not an end in itself. Personal safety depends on how researchers handle these tools and adapt to contingencies in the field. However, if deployed sensibly, based on a thoughtful threat model and as part of a digital-security strategy, they go a long way in protecting researchers and their partners from digital surveillance. It is important to stress at this point the fundamental difference between strategies and tactics: a strategy is broad and complex, comprising several tactics; tactics, in turn, combine a mix of different practices. It is at the level of practices that discrete digital tools, software applications, and hardware hacks come into play. To devise an efficient tactic, usually several tools need to be combined.

## Communicating safely

Most communication technologies should be treated as unsafe in principle. For mobile phones and their corresponding applications, this is particularly true (more than, for example, for desktop or laptop computers).<sup>1</sup> Researchers' location and movements can be

easily tracked via cell phones, and their phone calls or text messages are easily intercepted by state agencies. Fake cell-phone towers, so-called IMSI (International Mobile Subscriber Identity) catchers, are routinely used by police and intelligence agencies across the globe to intercept phone calls by target persons.<sup>2</sup> Furthermore, it has become a routine practice for law enforcement officers, border control agents, and even non-state actors, such as militias or cartel members, to seize and search researchers' mobile phones at checkpoints, border crossings, and even at conferences, for clues about their intentions and networks.

Nevertheless, in many field sites it is not only more convenient to rely on a smartphone, instead of carrying a laptop, but it is also safer given the lower visibility in public places. In such circumstances, it is worthwhile to prepare for random searches of your phone by controlling what content is visible in your messaging apps and contact lists during a superficial sweep. Many messengers now allow users to delete conversations or partial chat records not only from their device, but also from that of their interlocutor. These messages are not actually deleted from the application's servers, but they no longer show in the phone record and will not raise suspicion during a superficial scan. Furthermore, you can make your mobile phone communication much more secure if you rely on software with end-to-end encryption, that is, a system where only the communicating users, not telecom providers or internet service providers, can read the messages. *Signal*, an open-source messaging app, was once novel for providing this encryption feature while offering a user-friendly interface. Now, commercial tools such as *WhatsApp*, *Viber*, and *Telegram* have also adopted this encryption protocol, making a relatively secure way of communicating available to millions of inexperienced users. However, while the content of messages sent through these messaging applications is secret, their metadata is not. Data activists have pointed out that this is particularly contentious since it is commercially owned and closed source code software (Kazansk et al. 2016). The number and length of voice calls and messages, the volume of transferred data, their sender and addressees, and the timestamp of messages is exposed to the service providers, and thus potentially also to other third parties.

Particularly in the case of *WhatsApp*, which was bought by *Facebook* in 2014, several instances are documented where user data was sold to third parties or subpoenaed by law enforcement agencies.<sup>3</sup> As Van der Velden stresses, individual users will of course 'rarely learn that governments requested access to their data, as the disclosure of this information is supervised in secret court rulings' (Van der Velden 2015, p. 193). Additionally, most encrypted commercial messaging apps (including *Signal*) use phone numbers as identifiers for users. This effectively means that your identity is tied to your usage of the app, and the same is true for all your interlocutors in the field.<sup>4</sup> In addition to concerns about source

<sup>1</sup>For an impressive example of what can be learnt from only the mobile usage data of an individual tracked over a period of six months, see: [www.zeit.de/digital/datenschutz/2011-02/vorratsdaten-malte-spitz](http://www.zeit.de/digital/datenschutz/2011-02/vorratsdaten-malte-spitz).

<sup>2</sup>For a description of IMSI catchers and how they work, see: <https://github.com/b-meson/securitytalks/blob/master/IMSICatchersForActivists.md>.

<sup>3</sup>Verizon lists numbers of subpoenas for user data issued by the US state between 2015 and 2017 at: [www.verizon.com/about/portal/transparency-report/us-report/](http://www.verizon.com/about/portal/transparency-report/us-report/).

<sup>4</sup>*Wire* is based on *Signal*'s protocols but does not require users' phone numbers, which is a significant advantage since in many countries it is hard or impossible to disconnect your phone number from your identity.

protection, in practice, researchers may have many personal and professional reasons not to share their private numbers with their field contacts. Many experienced field researchers have, for instance, described the risk of sexual harassment by field contacts as ‘something familiar’ (Glasius et al. 2018, p. 31). In particular, solo-traveling female researchers might be hesitant to disclose their phones number to unknown field contacts (York 2017). One option to mitigate this problem could be the use of an additional SIM-card with a new number for fieldwork, or a burner phone, that is, a cheap second phone that is used only for communication in the field and can be disposed of if needed.<sup>5</sup> Generally, researchers should be aware that using the same phone to call or text multiple interlocutors has the consequence of constructing a bridge between them and creating a so-called social graph of the user. Their correspondence with field contacts thus inevitably turns the latter into a target group for surveillance which, prior to profiling based on telecoms data, may not have existed.

Digital-security trainers recommend computer chat and mobile messaging encryption based on OTR (‘off-the-record’) or similar protocols, such as *ChatSecure*.<sup>6</sup> This software, above all, has the advantage of enabling deniability. After a chat session has ended, the origin and destination of sent messages cannot be identified. This means that messages cannot be positively traced back to researchers or their correspondents. Even if third parties obtain the private identification keys of either a researcher or her/his correspondents, previous conversations cannot be compromised. While considerably more secure than other messaging, the downside of OTR messaging is that it is limited to synchronous communication: both correspondents must be online at the same time in order to converse. Hence, OTR-reliant software might provide a good platform to switch to with an interlocutor when intending to convey sensitive information. To establish first contact and for everyday messaging, however, its uses are limited (only a few of our field contacts are currently using OTR technology).<sup>7</sup>

### Box 6.1

#### Off-the-record (OTR)

OTR messaging is a secure messaging and chat protocol that offers the advantage that both ends of the communication can be assured that their correspondent is really who he/she claims to be, provided that they both have authenticated each other. Authentication can be conducted through a shared secret that they agree upon in advance, via a secret question and answer, or by comparing OTR fingerprints through a third person. The risk of human error or of researchers or their interlocutors being entrapped via social engineering is thus substantially reduced.

<sup>5</sup>Detailed guidelines for self-protection against online harassment and cyber-mobs are provided by the creators of Feminist Frequency and Women, Action at: <https://onlinesafety.feministfrequency.com/en/>. Hackblossom’s illustrated protection guide against stalking and digital harassment is also a useful resource for researchers: <https://hackblossom.org/domestic-violence/index.html>.

<sup>6</sup>See: <https://chatsecure.org/about/>.

<sup>7</sup>To set up OTR-based messaging on your devices you can follow the accessible technical instructions provided by Micah Lee for *The Intercept* at: <https://theintercept.com/2015/07/14/communicating-secret-watched/>.

Encrypted communication is not limited to mobile phones; however, the only reliable way to encrypt your email is via *Pretty Good Privacy* (PGP), an asymmetric encryption. Researchers need to assess whether they want to use PGP to protect their communication. It cannot easily be used on a smartphone, it is not integrable into many mail clients, and the principle by which PGP works is complicated and hard to explain, even by technically-versed trainers.<sup>8</sup> Moreover, PGP is difficult to use correctly, making it prone to human error. *Motherboard* reported recently that even the creator of PGP, Phil Zimmermann, has stopped using the technique because of its inconveniences (Franceschi-Bicchierai 2015). In some cases, using PGP might still be key. In others, however, it may be enough to suggest alternative communication means, which they may already be using, to research participants. For instance, it is a small step to switch conversations from *Facebook Messenger* or unencrypted email to a more secure and equally convenient messaging tool, such as *Signal*, *Wire*, or *ChatSecure*. The same principle applies to video-chat software: instead of relying on *Skype* (which participates in PRISM, thus sharing data with governments) for interviews or team meetings with fellow researchers and supervisors, researchers can make use of encrypted online video-chat platforms, such as *Jitsi Meet*.<sup>9</sup> The platforms run directly within the browser and the only thing participants need to access a group conversation is a conference URL. A researcher who wants to initiate a conversation simply creates a password-protected link on the platform and shares it with the desired participants.

## Secure data collection and processing

Although there are inherent vulnerabilities in digital communication technologies, you should not see computer software and mobile phone applications solely as enemies. If you choose digital tools for communication and field-data processing by whether their source code is open or closed, several threats can be mitigated systemically. With open source code, the chances are decreased that the software has backdoors for intruders or sends out some of the content it accesses to a third party. Free and open-source software (FOSS) is transparent, as it is based on open source code, which can be freely accessed. This means that every software developer, security researcher, or other technologically versed individual, can take part in auditing the software by scrutinizing its architecture. If bugs become apparent during this review process, they can be reported to the wider community and media. It takes only one software user who is committed to making sure that the tools she/he uses do not contain any vulnerabilities or malicious code to detect vulnerabilities and warn others. Such a collective and transparent source-code audit is thus an effective way for evaluating the security level of a software tool.

With closed-source software, only the developers and their contracted auditors have the means to detect weaknesses and ultimately know what a software program really

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<sup>8</sup>For a video tutorial about public-key cryptography and how it works, see: <https://vimeo.com/132517596>.

<sup>9</sup>See: <https://meet.jit.si/>.

does. Unlike commercial service providers, whose business model is based on monetizing data, FOSS usually originates in collective projects that are based on a shared ethical framework for digital technologies, which strongly emphasizes privacy concerns. Popular free and open-source operating systems that have been developed this way are *Debian*, *Ubuntu*, *Subgraph OS*, *Qubes OS*, and *The Anonymous Incognito Live System (Tails)*.

Researchers can draw on an entire range of such tools as better alternatives to insecure commercial ICTs. Not all are sophisticated and require deep technical knowledge. Many have a low access barrier in addition to being free, making them useful replacements for commercial software. In academic institutions, there is a clear advantage to working with FOSS, because it is free to access, to use, and to modify according to the specific needs of users.<sup>10</sup> However, setting up this infrastructure, becoming familiar with it, and maintaining it requires time, skills, and resources (especially for researchers who are already accustomed to commercial software).

## Box 6.2

### Privacy-by-design

Free and open-source software tools, such as *Tor* or *ChatSecure*, are built according to the privacy-by-design principle. Privacy-by-design shifts trust from commercial service providers' promises to the transparent architecture of a tool. This means that users do not need to trust service providers not to share their data with undisclosed parties or cooperate with potentially hostile authorities. Researcher using FOSS can instead rely on a community of information-security researchers and developers on the lookout for malicious code.

### Browsing safely

One useful and easily accessible alternative to commercial software that caters to the needs of field researchers is being developed by the *Tor Project*.<sup>11</sup> The *Firefox*-based *Tor Browser* is an invaluable tool for researchers of hostile environments and an alternative to insecure browsing software, such as *Google Chrome* or *Mozilla Firefox*. The *Tor Browser* comes with a default HTTPS setting (see Box 6.3), providing reasonable assurance that researchers are actually accessing the website that they intended to access (as opposed to an impostor), without attackers eavesdropping or otherwise interfering.<sup>12</sup> Due to its nodal architecture, which protects users' IP addresses, it provides anonymity and privacy for web queries.

<sup>10</sup>For useful suggestions of open-source alternatives to popular proprietary applications, see: <https://osalt.com/> and the Courage Foundation's <https://prism-break.org/en/>.

<sup>11</sup>For further information on the *Tor Project* and its software releases, see: [www.torproject.org/index.html.en](http://www.torproject.org/index.html.en).

<sup>12</sup>For a visualization of what *Tor* and HTTPS protect against, see: [www.eff.org/pages/tor-and-https](http://www.eff.org/pages/tor-and-https).

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### Box 6.3

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#### *Tor*

The *Tor* network is a group of volunteer-operated servers. Internet connections are via a circuitous route through a series of virtual tunnels, rather than directly. In a nutshell, different devices become nodes in a network, and information passes through this network via several detours. Each node communicates only with the prior sender and subsequent destination; only the first node knows where the request originated; only the last one knows where it is going. Not a single node knows both the initial sender and the addressee. This system allows users to maintain anonymity while browsing and to protect the privacy of their communications from service providers and authorities.

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*Tor* is useful for all stages of field research. Before going into the field, researchers can use the *Tor Browser* for desk research, to mitigate profiling through analytics (see the section on Thinking of Data Holistically, in Chapter 5), which reduces the risk of revealing sensitive information about your research, such as your interlocutors' identities. In the field, it can provide some protection to researchers who have either not set up email encryption or need to correspond with friends, supervisors, or interlocutors who are not using PGP. If they access their webmail accounts through the *Tor Browser* or route their connection to their university servers over *Tor*, the communicated content might not be encrypted, but security is significantly improved by the architecture of the *Tor* network and the security settings of the *Tor Browser* (including *NoScript*, *HTTPS Everywhere*, *JavaScript* disabled on non-HTTPS sites, ad blocking, and only temporary cookie storage). Significantly, *Tor* also enables censorship circumvention and reaching otherwise blocked destinations or content. This feature is particularly helpful for field sites where state actors are blocking access to certain websites or software.

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### Box 6.4

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#### HTTPS

Hypertext Transfer Protocol Secure (or HTTPS) is an adaptation of HTTP for secure communication over a computer network. It creates a secure layer over the insecure HTTP network. In the underlying HTTP, the accessed website (host), the IP address of the person accessing it, and port numbers of web servers are not protected. Attackers cannot see the content of communication, but they can infer the user's address, the duration of the communication, and the amount of data transferred. With the help of the plug-in *HTTPS Everywhere*, conventional web browsers can be made substantially safer, as they are routinely routed to HTTPS versions of accessed websites. For further information and a secure download link to the plug-in, see: <https://www.eff.org/de/https-everywhere>.

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On *Android* smartphones, researchers can also install *Tor*-based applications for mobile browsing, such as *Orbot* and the *Firefox*-based *Orfox*, which route their web queries and

traffic through the *Tor* network.<sup>13</sup> Some messaging applications, such as *Signal* or *Wire*, can be routed through *Tor* with the help of *Orbot*. While this does not work for voice chat, only for messaging, it may come in handy in situations where the central servers of a messaging service provider are blocked (e.g., for censorship circumvention purposes), and for situations where researchers do not wish to reveal to their phone-service providers that they are using a certain messaging app. Researchers should, however, be aware that this feature is also the main reason why *Tor* has been banned in several authoritarian contexts (e.g., in China, Turkey, Iraq, and the United Arab Emirates). Before routing internet traffic through *Tor*, researchers should familiarize themselves with the legal ramifications and, at the very least, closely track security incidents related to the use of *Tor* (and any other software).

## Entrapment and phishing

When we focus on the digital aspects of risk, we have to remember that older forms of threats (including intelligence agents and postal interception) are still a consideration. In fact, the old and new threats often come in pairs. Social engineering is a perfect example of this. These tactics use socio-psychological manipulation in combination with digital technologies to obtain access to (sensitive) data. For example, an attacker can call your online service provider pretending to be you, with the goal of getting customer service or technical support to share one piece of data that can then be used to unlock further data. A common social-engineering attack, and one that has been used to target activists in hostile environments, is called *phishing*. This is how it usually works: you receive an email, instant message, or private message on *Twitter* or *Facebook* from either an unknown person or a contact who poses as somebody you know. In this message, you are invited to open an attachment containing malware. Alternatively, you are asked to click on a link, which leads to a login page masquerading as *Dropbox*, *Outlook*, *Google*, or other familiar online services. On this fake portal, which often closely resembles the legitimate original, you are then prompted to provide username and password, or other sensitive data.<sup>14</sup>

While some of these phishing attacks are obvious, others can be quite convincing, building on open-access information about your current work and coming from a friendly source. One of our workshop participants recounts his experience working in an Egyptian NGO that was targeted by a phishing attack – an incident that came to be known as the *Nile Phish* attack. In this concerted campaign, over 90 often highly personalized messages were sent to members of seven reputable human-rights organizations and a number of well-respected lawyers, journalists, and activists, inviting them to take part in a panel discussion on a new draft NGO law, which had been in the making at the time.

<sup>13</sup>More information on *Orbot* and *Orfox* are provided by the Guardian Project at <https://guardianproject.info/apps/orbot/> and <https://guardianproject.info/apps/orfox/>.

<sup>14</sup>For more information, the *Freedom of the Press Foundations* Anti-Phishing and Email Hygiene guide covers common phishing tactics and how to avoid them: <https://freedom.press/training/email-security-tips/>.

The emails appeared to come from the *Nadeem Center*, a reputable anti-torture organization, and included a link to a website asking them for their password to view further details about the panel meeting. Different versions of this phishing attack have been reported since November 2016 and are being investigated by *Citizen Lab* and others.<sup>15</sup> Similar targeted phishing attacks (spear-phishing) affect researchers as well, often without their knowledge. For instance, many scholars have received spam invitations from supposed journal editors, asking them to join an editorial team, provide their services as an academic reviewer, or submit an article. What some researchers are unaware of is that some of these emails are effectively phishing attempts that set their trap with an enticement that a scholar might go for.

There is a lot you can do to avoid falling for such honeypots. Generally, think twice when prompted to provide any sensitive information on a web form. The best password or encryption key does not help if you are willing to enter it incautiously in any online form. Setting up two-factor authentication for all digital services, especially your email account, allows for better account security.<sup>16 17</sup> Moreover, it provides some protection, should login information fall into the wrong hands. What is more, if a link or an email looks ‘phishy,’ it may be better to forward it to your IT department before opening it. It is also sensible to alert colleagues and the supposed message sender of the phishing attempt. The ‘sender’ may be unaware of the abuse of his/her address – and, he/she might be quickest to corroborate or rebut your suspicions.

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### Box 6.5

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#### Accessing URL

One of the most important ways to improve your digital security with little effort is by avoiding opening links directly from email programs. This includes unsolicited links or attachments from familiar senders, as their computers may be compromised. In December 2016, the staff of the German think tank *Stiftung Wissenschaft und Politik Berlin (SWP)* received an email from the institute’s bookkeeper regarding their Christmas bonuses with an attached Trojan virus disguised as an excel file. To avoid falling prey to such phishing attempts, save the attachments to a folder and open it only after scanning it with antivirus software. You should also avoid following hyperlinks in email-account notifications; instead, log in to the respective service directly, using a bookmark or the websites <https://> URL. Examine URL links before activating them, not only when you have been unexpectedly directed to a login screen or redirected to an unfamiliar warning page. In a similar vein, it is much safer to copy and paste links sent in emails into a secure browser, such as *Tor*, which prevents the destination page from running malicious scripts. Researchers can check whether a URL or file is malicious at [www.virustotal.com/](http://www.virustotal.com/).

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<sup>15</sup>For more information on the Nile Phish case, see: <https://citizenlab.ca/2017/02/nilephish-report/>.

<sup>16</sup>For a useful list of websites that support two-factor authentication, see: <https://twofactorauth.org/>.

<sup>17</sup>For a useful list of websites that support two-factor authentication, see: <https://twofactorauth.org/>.

## Protecting stored data

When it comes to securely storing data, encryption is generally superior to non-encryption. However, different devices require different disk-encryption techniques, and those techniques vary significantly in their ability to protect research data. On a general level, disk encryption is a technology that protects data stored in a device by encoding it to an unreadable (encoded) format. In order to get a readable (decoded) format of these data, you need access to a security combination – usually a passphrase or a cryptographic key, or both. This is a useful way to prevent unauthorized access to stored data.

The major types of disk encryption are full-disk encryption and filesystem-level encryption. The former encrypts all data on the device, including the metadata that could otherwise give hints about the encrypted file content. Some operating systems, such as the Linux-based *Ubuntu*, *Fedora*, *Debian* and *Qubes*, *Microsofts Windows Pro*, and *Apples iOS*, come with a full-disk-encryption option upon installation – but differ significantly with a view to the security thereby provided this way.<sup>18</sup> Others need additional software. In addition, the latter encrypts only specific files or directories on a device. Access to these files is thus barred, although filesystem-level encryption usually does not encrypt their metadata, such as file sizes, names, modification timestamps, or directory structure.<sup>19</sup>

Both encryption types have pros and cons and are most powerful when combined. Full-disk encryption is, above all, resistant to offline attacks, for instance, by an individual trying to get physical access to a switched-off device in order to copy all data from its hard-drives. However, once the device is successfully decrypted, all files and directories on the system are unprotected. This is also the case when users do not log out of their operating system when leaving their laptops, e.g., for a break. Moreover, users of proprietary software entrust the respective software company with storing their encryption key. In the case of the Windows-owned *BitLocker*, the encryption key is stored in users' *Microsoft* accounts (Lee 2015). These accounts are vulnerable to targeted hacking attacks or can be subpoenaed by state authorities. A strong filesystem-level encryption for individual files or folders, such as the one offered by the open-source software *VeraCrypt*, can prevent data from falling into the wrong hands.<sup>20</sup> The software not only protects files with a password, it also allows you to create a hidden volume within an encrypted file which only appears on the screen when a different pass phrase is entered. As this hidden volume is concealed in the free space of your encrypted folder, its existence should be almost impossible to detect for third parties. This nested encryption of files thus enables plausible deniability in situations where you simply cannot refuse to reveal your passwords, for instance, due to physical threats or extortion. Should you ever be forced to hand over your computer and the passwords to your encrypted volumes, your hidden files will still be safe.

<sup>18</sup>For a guide to disk encryption for Windows and Apple devices, and for Linux machines at the time of installation, see: <https://theintercept.com/2015/04/27/encrypting-laptop-like-mean/>.

<sup>19</sup>Ted Han and Quinn Norton provide advice how to 'clean' files of this unwanted metadata: <https://source.opennews.org/articles/how-protect-your-sources-when-releasing-sensitive-/>.

<sup>20</sup>See: <https://securityinabox.org/en/guide/veracrypt/windows/>.

In some fieldwork contexts, state agents may, in principle, have the IT-resources to break into encrypted devices. However, often these resources are scarce and not operable on a mass level. Officials who seize researchers' devices in the field may also not even have access to these resources. Albeit not bulletproof, file encryption thus generally poses an effective additional barrier against intrusion. A strong passphrase – a long password consisting of several random words – is essential for increasing the intrusion costs for attackers. Here it is noteworthy that the typical password requirements – a combination of numbers, upper- and lowercase letters, and special characters – are surprisingly insecure. As Micah Lee (2015) notes, 'coming up with a good passphrase by just thinking of one is incredibly hard, and if your adversary really is capable of one trillion guesses per second, you'll probably do a bad job of it.' In turn, although fingerprint protection is convenient, you may be easily compelled to unlock the device (Heisler 2014). A simple way to come up with an easy-to-memorize but very secure passphrase is the Diceware method.<sup>21</sup> This passphrase can then be used as a master password for password-management software, where all other passwords (e.g., to email accounts, shopping websites, university servers, file archives, etc.) are stored and copy-and-pasted when needed.<sup>22</sup> A second Diceware passphrase can become the key to your encrypted hard-drive for logging into your computer. This way, if one password is compromised, all the others stored in the password manager are still safe.

With a view to mobile phones and handheld devices, the use of such long random passphrases is admittedly impractical. Here iPhone users can configure the privacy settings so that stored information is deleted after a certain number of unsuccessful attempts to unlock it (UNESCO and Reporters Without Borders 2016, p. 76). While such a measure cannot prevent infiltration by digital means (e.g., via malware that is installed while the phone is online), it can mitigate the risk of contacts and chat protocols being compromised if the phone is stolen or seized. This substantially raises the protection level of a simple four-digit passcode, which is easy to crack by brute force attack. For Android devices, such technology is not yet built in. For these devices, protection against security breaches is offered only by the standard device encryption, which itself still depends on the strength of the chosen passphrase.

## Obfuscation and pseudonyms

Digital security is as much about creativity as it is about technology. One strategy that has proved useful and is easily accessible to researchers with even minimal technical knowledge is obfuscation. Obfuscation is the obscuring and concealing of messages through code language, hidden subtext, circumlocution, or jargon. Pseudonyms for respondents,

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<sup>21</sup>For details on the method, see: [www.eff.org/dice](http://www.eff.org/dice).

<sup>22</sup>Many of these secondary passwords for online services, such as social-media sites, do not need to be very strong, since each guess by a potential intruder requires communicating with a server on the internet and most service providers limit the number of unsuccessful guesses. For a discussion of random word combinations and examples of random wordlists, see: [www.eff.org/deeplinks/2016/07/new-wordlists-random-passphrases](http://www.eff.org/deeplinks/2016/07/new-wordlists-random-passphrases).

meeting points, and research tasks is an intentional obfuscation strategy that researchers use on a regular basis. The idea can translate to ICTs in research as well. For instance, you can obfuscate audio recordings of interviews before offloading them by renaming them and changing their file extensions in a way that makes them seem uninteresting. The file 'Interview1.mp3' may be renamed 'Holiday expenses.xlsx,' seemingly an Excel spreadsheet listing the expenses of your last holidays. As such, it might be overlooked by law enforcement officers doing a superficial sweep of a hard-drive or mobile phone. The modification of the file extension will also change the visible appearance of the file: the MP3 icon will turn into an Excel icon.<sup>23 24</sup> Reporters Without Borders offers similar advice for emails with sensitive content, recommending that you make such messages appear to be spam by giving them innocuous subject lines, such as Miracle Diet Offer (UNESCO and Reporters Without Borders 2016, p. 74).

Certain tools may assist the obfuscation of data. For instance, the messaging app *PixelKnot* provides a means to embed plain text as password-protected code in any mobile phone picture.<sup>25</sup> This enables researchers to hide sensitive contact information or even entire field reports in seemingly harmless holiday photos that will not raise suspicion in a random phone inspection at a checkpoint or border-crossing.

## Offloading

Data storage remains a key point for potential compromises, precisely because no encryption or obfuscation practice ensures complete security. Effectively offloading data might be the most important element of digital data management. The basic idea is to design a routing practice for transferring sensitive information as soon as possible off the hard-drives of any computer or device that a researcher carries into the field in order to minimize the timeframe during which the information can be seized. Glasius et al. (2018, p. 33) tell the story of a field researcher in Gujarat, India, whose luggage was stolen or potentially confiscated during an airport layover. The lost luggage included a computer containing sensitive (albeit anonymized) interviews with opponents of Narendra Modi. In such unforeseen circumstances, it can be a relief to know that there is no sensitive data left on a device that could be compromised if the device is stolen or seized.

Safely and efficiently getting rid of sensitive transcripts, pictures, analyses, or records without incurring the risk of data loss is a considerable challenge. As the example above points out, USB drives, CDs, DVDs, and SD cards have the disadvantage that they can be physically seized or simply lost. Moreover, it is tough to know how to handle memory sticks

<sup>23</sup>Of course, this does not help, if intruders simply look for the most recently used files. File properties can also be manipulated to obscure their creation date and place. However, if researchers are going to the effort of doing this, they are better off encrypting their files, which offers easier and better protection.

<sup>24</sup>Of course, this does not help if intruders simply look for the most recently used files. File properties can also be manipulated to obscure their creation date and place. However, if researchers are going to the effort of doing this, they are better off encrypting their files, which offers easier and better protection.

<sup>25</sup>See: <https://guardianproject.info/apps/pixelknot/>.

in crisis situations: should you try to hide them and risk them being found? Or should you try to discard them and risk data loss? In the past, many activists hid their data in the discussion forums or html code of crowdsourcing platforms like *Wikipedia*. Nowadays, a few offloading tools are available that serve the same purpose while providing better accessibility. Most researchers already use *Google Docs*, *Dropbox*, or other cloud-based services to compile their field notes online. While some of these technologies are more secure than keeping files stored on the laptop in a hostile field context, their trustworthiness is questionable. Even though the term ‘cloud’ might suggest otherwise, the uploaded files are not somewhere in midair, but are stored in a physical space on a commercial server. When considering the use of a cloud service provider, you should thus keep in mind that there are safer alternatives. Again, the risk depends on the capacities of those who would like to gain access to the data. But the rule of thumb should be to never store files anywhere unless they are encrypted.

## Remote servers

For some authors of this book, university-owned IT infrastructure has been an alternative to commercial cloud servers. Some institutions have their own servers with discrete hard-disk volumes assigned to individual researchers. Others provide free web space to their researchers and supervisors that can be used to store sensitive data. Usually, university servers can be accessed from the field via VPN to paste files containing sensitive information.<sup>26</sup> In contexts where establishing VPN connections is illegal or where connections are monitored, university servers can be securely accessed through protected third-party applications with end-to-end encryption, such as ownCloud or SpiderOak.<sup>27</sup> Created primarily for whistleblowers, digital dead drops are another possibility for digital infrastructure that can be set up and maintained by institutions to support their field researchers (UNESCO and Reporters Without Borders 2016, p. 74).<sup>28</sup>

Even better than a university server is your own remote server that you do not take with you on the trip but leave with someone trustworthy. Generally, researchers have devised a range of practices for offloading that rely on a trusted colleague and can serve as an alternative to cloud technology. The basic idea is to have someone outside the reach of the authorities in the field who continuously scans and saves submitted field data to a secure drive. Like check-in tactics in hostile environments, this type of peer-to-peer offloading strategy requires discipline and a strict offloading plan. Aside from this, there are different tools that can be used to facilitate such an offloading practice. With the help of the open-source software *OnionShare*, for instance, you can send your files to a trusted contact.<sup>29</sup>

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<sup>26</sup>For a basic guide to VPN, see *The Motherboard Guide to VPNs* at: [https://motherboard.vice.com/en\\_us/article/vvkzbd/the-best-vpns-ranked](https://motherboard.vice.com/en_us/article/vvkzbd/the-best-vpns-ranked).

<sup>27</sup>See: <https://owncloud.org/> and <https://spideroak.com/>.

<sup>28</sup>The open-source whistle-blower submission system SecureDrop (<https://securedrop.org/>), for instance, was created by the late Aaron Swartz to enable media organizations to securely accept documents from and communicate with anonymous sources. It is currently managed by Freedom of the Press Foundation. For private users, Reporters Without Borders suggests using shared mailboxes as dead drops.

<sup>29</sup>See: <https://onionshare.org/>.

Routing the data through the *Tor* network, *OnionShare* has proved a convenient and trustworthy tool for synchronous file transfers. However, for successful file transmission, both sender and addressee need to have *Tor* installed on their computers and need to be online at the same time. Another alternative is the use of etherpads, which is already common practice among activist collectives. Popular etherpad hosts include *Riseup* and *systemli.org*.<sup>30</sup> Recently, encrypted pads such as *CryptPad* have also gained popularity, as they offer relatively good protection while also providing text-editing options that are integrable to conventional office software.<sup>31</sup> Etherpads are practical and easy to access, although they are only as secure as their host server and only as trustworthy as their host organizations. For instance, the popular public etherpad *LQDN* is hosted in France, where there is currently a state of emergency in force that could allow authorities to access to the pad's contents through a digital search warrant or by seizing the server hard-drives. Here again, researchers must decide themselves, depending on their threat model and research context.

### Offloading with *Tails*

One of the few practices that have been recommended to us by several security trainers, regardless of research context, is using the operating system *Tails* for more secure communications (browsing, chat, and email) and for offloading. *Tails* is a free and open-source operating system that boots from a USB drive and can be used in parallel with your regular operating system on your computers. It leaves no traces, routes all data traffic through the *Tor* network and hides the location and identity of the devices it is used with. If your laptop is searched or seized, it will not reveal anything about the sensitive data communicated via *Tails*. Moreover, *Tails* lives only on a USB that can be easily discarded. Thus, the chance that sensitive information will be seized is minimized. Additionally, *Tails* comes pre-packaged with several applications for secure communication (an email client that supports PGP, a chat client using secure end-to-end encryption, and the *Tor Browser*).

## Box 6.6

### *Tails*

Using *Tails* for a limited time and for a specific purpose, such as sending away field research notes from your normal device, is a tactic that enables efficient offloading of data while keeping a relatively low profile. Generally, you have two options for using *Tails* for offloading. First, you can create a volume on the *Tails* USB that will persist beyond separate sessions and store files in an encrypted and password-protected manner. Data can be offloaded at any moment and the USB discarded. Alternatively, whenever you want to avoid storing sensitive files at all on a physical device that can be seized, stolen, or lost, you can set up *Tails* as a non-persistent volume. This way, once ejected, *Tails* forgets everything it was used for (files opened or

<sup>30</sup>See: <https://riseup.net/> and <https://www.systemli.org/en/index.html>.

<sup>31</sup>See: <https://cryptpad.fr/>.

uploaded, browsing sessions, etc.) and the computer forgets the use of *Tails*. In both uses, *Tails* resolves the problem of securely deleting sensitive data from devices. It also provides a more secure environment for data offloading, as all traffic is routed through the *Tor* network. One way to use *Tails* would be to take a photo (with a webcam or camera) of field notes after an interview and immediately upload or send it to someone trusted via a computer running *Tails*. After uploading is completed, the sensitive file is removed from the computer and the USB key with *Tails* is ejected. For more details, see: <https://tails.boum.org/>.

## Compartmentalization

You can also think about other options to fit your individual research context and personal workflow. Perhaps you may split information into pieces stored in different locations, so that if only one part is seized, it is of not much use. These pieces can then be reassembled on your return home. This strategy is known as compartmentalization. One of the compartmentalizing tactics used by many researchers consists of separating uses of various online services and tools – for example, using a certain email account or *Facebook* only for communication with family and private friends and never for any research-related communications. Creating and using multiple accounts and online identities is a good way to exercise autonomy in managing data; however, you might also try to avoid or minimize having different types of data (private photos, files from different professional projects, web-browsing history, sensitive research information, etc.) in the same ‘place.’ Having different hard-drives stored at different locations, or having an additional device (called an airgapped device) that is never connected to the internet before, during, or after the field can be useful too. Having two devices, however, also comes with tradeoffs. Glasius and her research team describe how they had collectively agreed on a policy to always bring two laptops into the field: ‘one for web browsing, emails, and so on and one second-hand laptop that never went online, but acted almost as a typewriter, for transcribing interviews’ (Glasius et al. 2018, p. 32). Neither device would have details about field contacts. While this procedure reasonably protected respondents from digital surveillance, it increased the risk of losing transcripts and effectively raised the researchers’ exposure at border controls. The risk of data loss could have been mitigated by an efficient offloading practice.

## Operational security

When thinking about protecting sensitive, digitally-stored information, researchers and their advisors should consider that to be accessed by third parties, it ‘doesn’t need to pass through the Internet’ (Eguren and Caraj 2009, p. 115). Digital and physical security are tightly knit and mutually corrosive (when not done well) or mutually supportive (when done well). Ideally, security is treated holistically, and considers the social and psychological vulnerabilities of individuals and their support communities, as well as the potential weaknesses of everyday practices. Despite increasingly sophisticated encryption technologies and obfuscation tactics, no device is 100% secure. When facing a situation where your personal safety is in question, such as attending an event with a significant police presence,

you might thus consider leaving your digital devices storing sensitive information with a person you trust.

In other situations, a strong emphasis on operational security can offset many of the limitations of digital-protection measures. For instance, security breaches are easiest when intruders can get a hold of your device while you are distracted, for example during the coffee break at a conference or in an internet café. Therefore, security trainers advise researchers to always keep an eye on their laptops or smartphones. If attending a meeting where carrying hardware is deemed a security risk (e.g., because the microphone could be used to eavesdrop on a conversation), you may either disable the device by taking its batteries out or think of alternative ways of storing it. Removing a cell phone's battery is also a good way to avoid being tracked. Mobile phones communicate with cell towers, and these towers can track devices passively with surprising accuracy (and without GPS). Accordingly, mobile-network operators and potentially law enforcement agencies can 'see' where devices are, where they have been, and who they are near. Smartphones are thus always a potential 'snitch in your pocket' (Isikoff 2010). Removing the battery from a phone results in the device disassociating from the network, effectively protecting against this threat.

Many laptops do not have removable hard disks, however, and many smartphones no longer have removable batteries. With such devices, you cannot know if they are turned off, as malware can misinform about the power state (Silkie and Kamphuis 2016, p. 64). Preparation is thus essential. Think about which hardware is appropriate for your specific fieldwork context. This hardware naturally should be cleaned of any sensitive information before used in the field. To delete data from removable storage (e.g., an SD card or USB stick), you can perform a factory reset. Such a reset does leave data fragments on the flash storage, but they will be accessible only via sophisticated data forensics. For the same reason, it is not enough to delete files from computers by erasing them from the trash bin. Deleted data must also be 'wiped' – that is, overwritten – with specialized software. Ideally, after sensitive research processes, researchers could wipe their entire disk and just install a 'clean' new operating system. Alternatively, researchers could also get a cheap second laptop for the sole purpose of fieldwork. Furthermore, researchers may bring a flash drive with *Tails* to the field in order to have a portable, anonymous operating system on any computer they encounter. You could also get multiple pre-paid SIM cards and burner phones before traveling to a field-research destination and use them to separate communication with interviewees. This way, those who have not been previously connected do not get connected as a result of your fieldwork.

Increasing awareness to surveillance vulnerabilities entails also scrutinizing weaknesses with your hardware and software. Often these weak spots involve unconscious everyday behavior. The greatest password does not help if it is typed in front of an airport camera; a maximum-privacy browser, which leaves no traces of the viewed content on your hard-drive, cannot prevent those sitting behind you from seeing your screen. Moreover, when you change the software you use, take digital-security training, or alter how you use ICTs, it is imperative to think also about how this might affect other ways someone could target you. The sudden transition from communicating via landline to using a secure texting or telephone program may raise red flags where previously there had been none (Electronic Frontier Foundation 2018a). Learn and get used to everyday digital hygiene practices,

such as logging out of email and social-network accounts, making use of private browsers for your daily web queries, etc., before you engage in fieldwork (Silkie and Kamphuis 2016; UNESCO and Reporters Without Borders 2016, pp. 71–72). Digital-security routines demand a certain acclimatization period. You can make use of the time you are waiting for fieldwork approval to familiarize yourself with the idea of digital security as an inherent part of your work by rethinking some of your social routines and by practicing it at home. This way, you will be better prepared once you arrive in the field and less overwhelmed by having to maintain a digital-security framework.

## Conclusion

The aforementioned examples show how some risks posed by increasing reliance on digital technologies can be reduced with moderate effort. Researchers do not necessarily have to rely on secure technologies 24/7.<sup>32</sup> Sometimes it is not only easier but also sufficient to use them as needed, for a specific purpose or a given time (i.e., during fieldwork). Digital strategies need to be contextualized given the security threats and protection needs that arise from the specific research scenario: ‘You don’t need to buy some expensive encrypted phone system that claims to be “NSA-proof” if your biggest threat is physical surveillance from a private investigator with no access to internet surveillance tools’ (Electronic Frontier Foundation 2018a). If the primary threat in the field stems from local militias with little access to sophisticated spy technology, then it might make more sense to spend the little time and resources that are usually available to researchers on insurance policies and hostile environment awareness training (HEAT). When conducting fieldwork in states run by repressive governments that criminalize some secure technologies, you might want to avoid creating a pretext for law enforcement agencies to come after you. Rather than risking prosecution, in some field settings it can make sense to go back to classic note-taking and reduce digital exposure. Generally, seeking out local knowledge and learning from local security practices is likely to lead you to appropriate tactics that you might not otherwise have thought of. Due to their longstanding experience, local activists and researchers often have a good sense of where the risks are and have often established reliable practices.

Digital-security planning is also about managing expectations: no data handling practice or software tool can provide absolute protection. Researchers can, however, mitigate *most* expected threats related to their communication practices and their handling of research data with the help of the tactics and tools presented in this chapter. The majority of digital threats faced by researchers are low-scale attacks that can be easily fended off. For instance, some of the discussed cryptographic protocols (*OTR*, *Tor*) are still unbroken, to the best knowledge of digital-security experts, making them quite a safe bet; the use of end-to-end encryption and non-traceable communication tools will make it hard or even impossible for authoritarian regimes or other third parties to eavesdrop on researchers; finally, sensitivity to phishing, a good offloading strategy, and awareness for operational

<sup>32</sup>But, arguably, handling digital security becomes less complicated when you do not switch too often between different technologies.

security can effectively prevent third parties from accessing sensitive content from their computers. With little effort, researchers can achieve a lot – if they are only willing to try.

## Guiding questions

### Questions for researchers

Framework:

- Have you evaluated your options for mitigating data-related risk with the tactics and tools presented in this handbook?
- What are you willing to do in order to mitigate this risk? Are you prepared to invest the necessary time and resources and have you included this learning period into your research plan?
- Have you devised a strategic mix of tactics to protect your communication and your stored data? Here it is important to think about two levels: risk prevention and reaction strategies.
- Now that you have your strategic framework, have you consulted enough step-by-step instructions to implement it?
- Do you possess sufficient knowledge about the use of protection software and technical tools? What do you have to look up or learn?
- Have you talked about all these steps with your colleagues? Are they sensitized for digital risks and willing to support each other?

Offloading and compartmentalization:

- Have you thought about your digital exposure and its consequences for your research? Are you engaging with social media and could this increase the risks for yourself and your research participants?
- What is your offloading strategy for research data?
- Are you cleaning your devices/hard drives directly after processing and offloading sensitive files so you do not leave traces?
- Are you using commercial products to which safer free and open software solutions exist?
- Are you encrypting your research data right away?

Operational security:

- Could your devices be compromised? Have they been left unattended or handled by other persons?
- Are you carrying a device that could track your movements?
- Is your hardware and software up to date? Are there maybe more secure alternatives?
- Which of these technologies do you really need in the field and which are best left at home?
- Are you sensitized to identify phishing attacks?

## Questions for advisors/supervisors

### Framework:

- Are you providing enough time and space to your supervisees to engage in detail with the topic of digital security before enabling them to engage in field research?
- Have you considered hosting digital-security workshops for your team members and supervisees, or inviting a digital-security expert to join your research team?
- Have you thought about including data security training in your teaching curriculum? Would you consider advocating for the systematic inclusion of such courses into the mandatory methodology classes?
- Have you talked about the topic of digital security and the above steps with your colleagues? Are they sensitized for digital risks and willing to support each other?

### Offloading and compartmentalization:

- Are you keeping an eye on the online exposure of your colleagues and supervisees before and after the fieldwork?
- Have you considered designing an offloading strategy for sensitive data with your research partners or supervisees?
- Are you supplying or could you supply the required software for better data protection, including commercial licenses, if needed?
- Could you take care of better IT-support through either departmental or private channels?

### Operational security:

- Are you supplying or could you supply the required hardware for better data protection, including, for instance, specific laptops that are dedicated to field work only?
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